# Grid Computing for Belle II Analysis

## Certificates · VOMS · gbasf2 — hands-on with the new IHEP CA

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# 1 · The Grid & the New CA

# What is the Belle II grid?

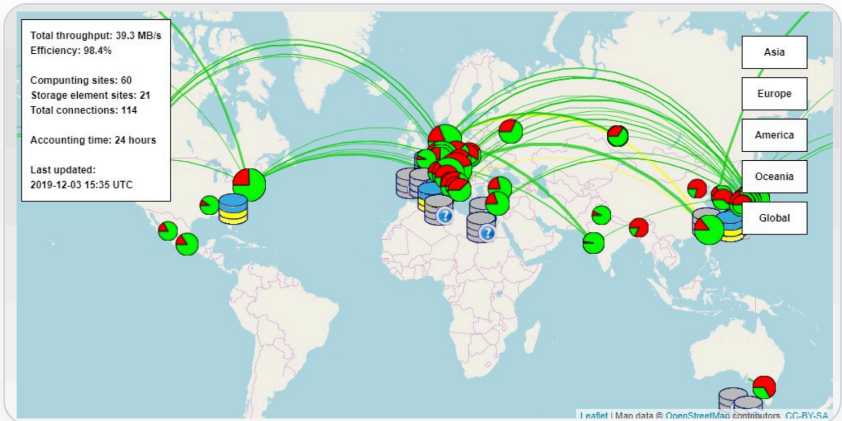
> **Note**
> The grid is a **distributed computing system** used by Belle II — and many other experiments — to harness computing resources of universities and institutes worldwide.

## Why not one big data center?

- Modern experiments collect **tens of petabytes**
- Processing all data / MC at a single site is **not sustainable**
- Belle II runs on grid sites across Asia, Europe and America

## The grid is used for:

- MC production · data reprocessing · skimming
- **Your analysis** — this is where you come in



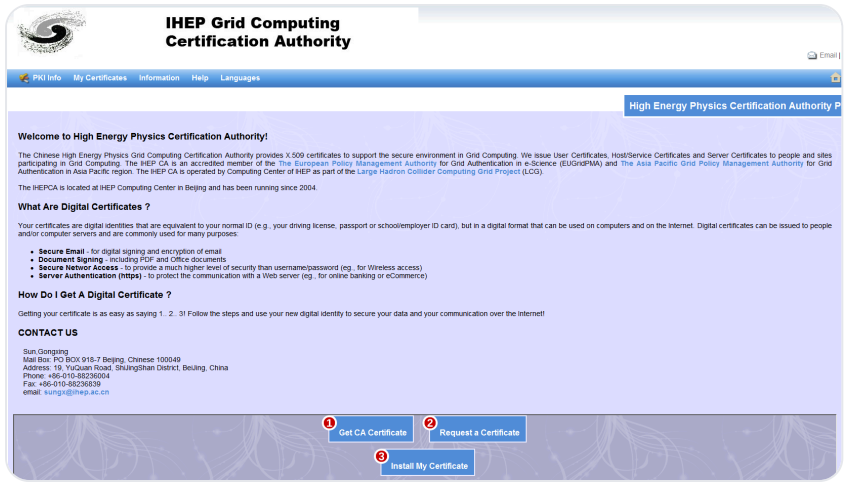

Grid sites and transfers worldwide — computing centers, storage elements and their connections.

View the live map at belle2.jp/computing (opens in your browser).

**Your path today:**

certificate → VO membership → proxy → gbasf2 jobs

# What changed: OpenCA → OpenXPKI




cagrid.ihep.ac.cn — the old CA web interface

| Dimension | Old (OpenCA) | New (OpenXPKI) |
|---|---|---|
| URL | cagrid.ihep.ac.cn | **gridca.ihep.ac.cn** |
| Login | Basic web form | **IHEP Account** + cert migration |
| Approval | Offline email | **Online RA workflow** |
| Key generation | Browser-dependent | **Server-side, RSA 4096** |
| Automation | None | **EST · API** |
| CRL | Manual | **Automatic** |

> ⚠ **Warning**
> Old CA certificates will **no longer be recognized after end of 2026** — about 45 days after this school. Migrate or re-apply **now**.

# 2 · Before You Start

# Preparation checklist

### IHEP Account / LDAP login works

You will log in to the CA WebUI with your usual IHEP username and password (IHEP Account, formerly IHEP SSO). Test it at login.ihep.ac.cn first.

### Your LDAP email is correct

It becomes the certificate's email SAN, and the system uses it for ownership and notifications. Wrong email → lost certificates.

### Decide your English name spelling

The certificate Subject needs ASCII letters only (space, dot, hyphen, apostrophe allowed). Use the same spelling forever — renewals must match.

### Prepare an export password

A secret only you know, used to encrypt the downloaded `.p12` file. It is **not** your LDAP password and cannot be recovered by the CA.

**Note**

Open **https://gridca.ihep.ac.cn/webui/ihepca/** in a modern browser (Firefox recommended). The WebUI, EST and login all require HTTPS.

# Three certificate types

| Type | Purpose | Private key generated & held by | How to obtain |
|---|---|---|---|
| **User** | Grid user authentication — browser / CLI clients | CA generates; you download `.p12` and protect it | WebUI |
| **Robot** | Non-human identities: experiment services, pilots, automation | CA generates; the responsible person downloads `.p12` | WebUI |
| **Host** | Server authentication for machines | CA generates (WebUI first issue); **EST auto-renews on the host** | WebUI + EST |

**Note**

Not sure which one you need? As a student doing analysis: **User Certificate**. Running a Belle II pilot service or an automated production task: ask your group leader about a **Robot Certificate**.

# What is inside a certificate

| Information | Where it lives | Can you change it? |
|---|---|---|
| **Subject DN** (name, account, OU, O) | The certificate itself | No — keep identical across renewals |
| **SAN** (email / DNS names) | The certificate itself | No — changed DNS or ownership ⇒ new application |
| Issuer · CRL URL · Certificate Policies | The certificate itself | No |
| Requester · Owner email · Comment | OpenXPKI system metadata | Only via workflow / RA |

## Your real certificate, as openssl shows it:

```
subject=C=CN, O=HEP, O=IHEP, OU=CC, CN=hanx14, CN=Xiao Han
issuer=C=CN, O=HEP, CN=Institute of High Energy Physics Certification Authority 2026
```

> **Note**
> In slash form — used by VOMS and older grid tools — your DN is: `/C=CN/O=HEP/O=IHEP/OU=CC/CN=hanx14/CN=Xiao Han`. Grid services map it to VOMS roles and DIRAC accounts: **keep your name spelling forever**.

# 3 · Hands-on: User Certificate

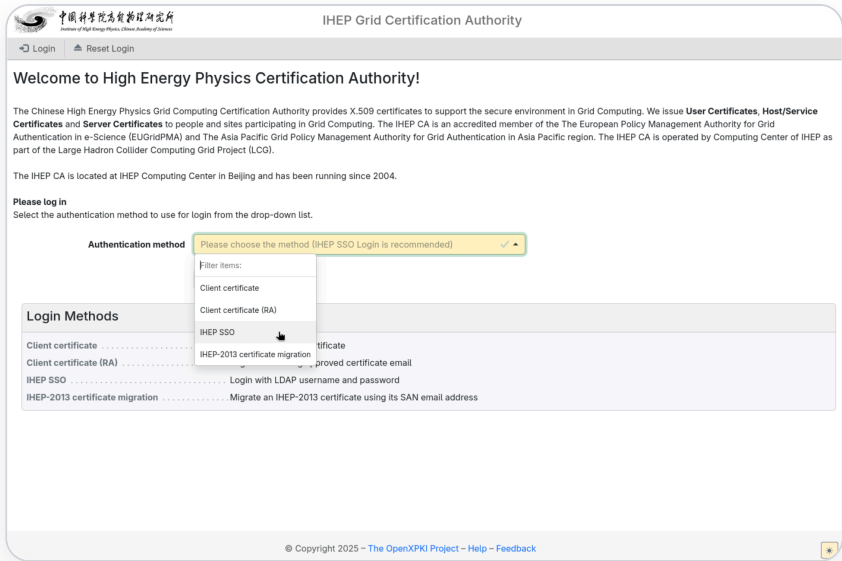
# Step 1 — Log in with IHEP Account

1. **Open the WebUI**
   https://gridca.ihep.ac.cn/webui/ihepca/

2. **Choose authentication method**
   In the dropdown select **IHEP Account**.

3. **Enter IHEP credentials**
   Username + password, then **Login**.

4. **Verify your account**
   Top-right corner must show **your** account. Logout via the button — not by closing the tab.




The WebUI login page — pick **IHEP Account** in the Authentication method dropdown.

> ⚠ **Warning**
> `Login with the given credentials failed!` → check LDAP account/password, do **not** brute-force different accounts; contact CA support to check the LDAP mapping.

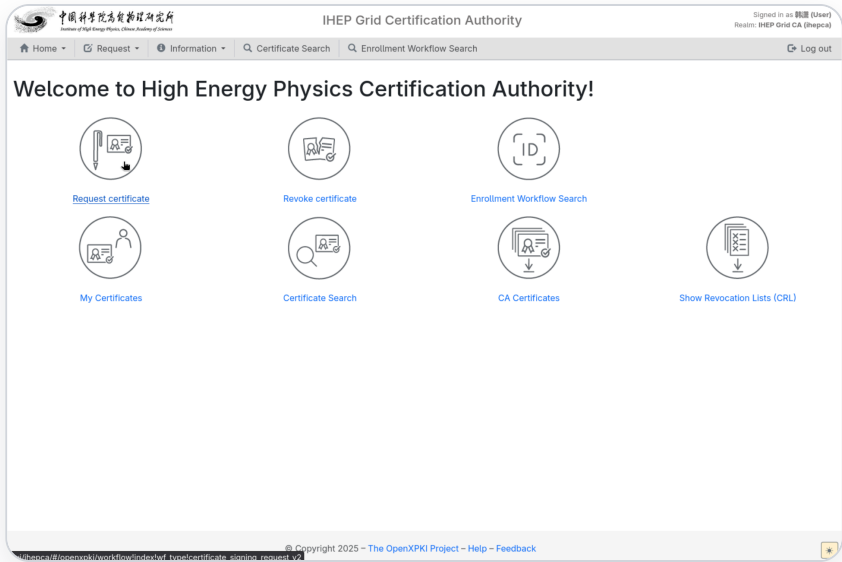
# Step 2 — Know your two home pages

**Home → My Certificates**

- All certificates owned by your account
- Renewal and revocation start from here
- Download actions live in the certificate detail page

**Home → My Workflows**

- Every request you created
- Status, history, and error messages
- Bookmark the workflow ID before contacting support




After login — the top menu with **Request Certificate**.

> **Note**
> These two pages answer 90% of "where is my certificate" questions. If a certificate is not in **My Certificates**, check which account (email) you logged in with.

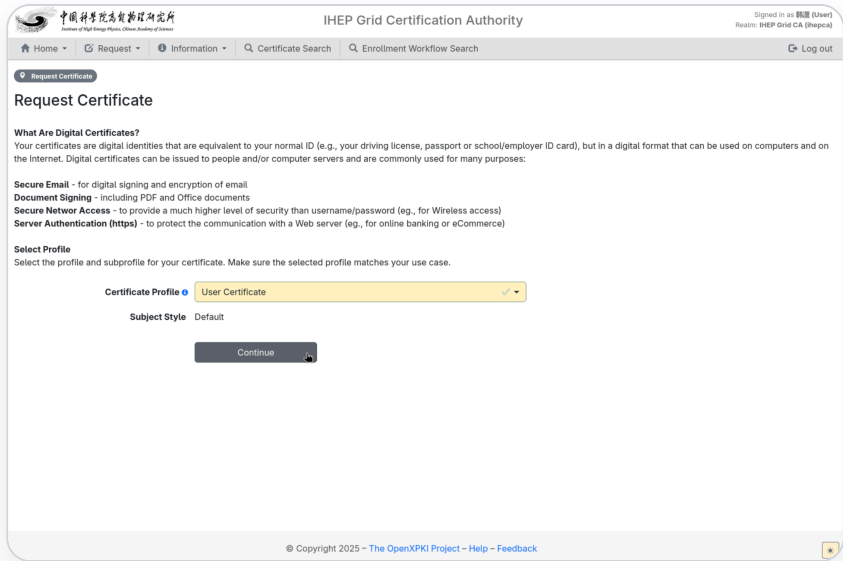
# Step 3 — Start a request

1. **Top menu → Request Certificate** (**Request → Request Certificate** in older menus.)

2. **Select Profile: User Certificate**
   It is the default — click **Continue**.

**The other profiles in the list:**

- **Robot Certificate** — for experiment services and pilots (§7)
- **Host Certificate** — for machines (§7)




**Select Profile** — User Certificate is the default.

> **Note**
> **Continue** only saves and moves to the next page. **Submit** is what actually sends the request to the system and RA. Review the summary before every Submit.

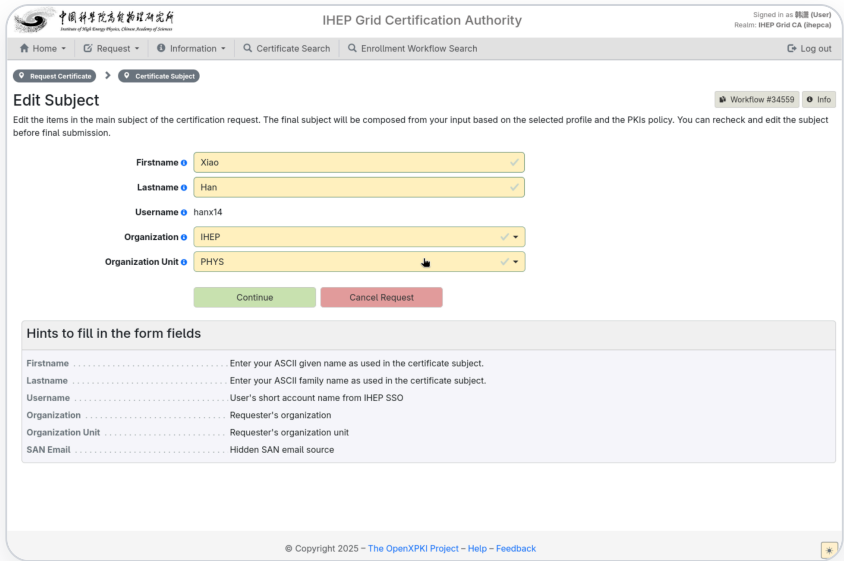
# Step 4 — Edit Subject

1. **First name / Last name**
   ASCII only: letters, space, dot, hyphen, apostrophe. If LDAP returns your Chinese name, **type your English name yourself**.

2. **Username**
   Pre-filled from LDAP, locked — it binds the cert to your account.

3. **Organization Unit / Organization**
   Dropdowns — select your real department and institute.

4. **SAN Email**
   Provided by LDAP — leave it untouched.




**Edit Subject** — type your ASCII name; Username and email come from LDAP.

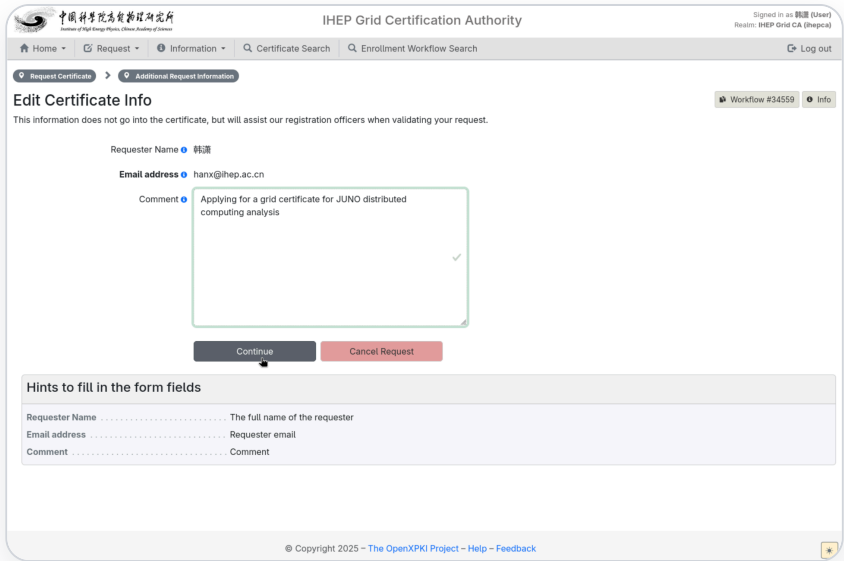
# Step 5 — Certificate info

1. **Email address**
   Your LDAP email — the ownership and notification address.

2. **Requester**
   Your name, as the applicant.

3. **Comment**
   e.g. "Belle II analysis user, winter school". **Never** put passwords or private key info here. Then **Continue**.

> **Note**
> Nothing here changes your identity — identity fields come from LDAP. Comment is free text for the RA reviewers.




**Edit Certificate Info** — write your application reason in Comment.

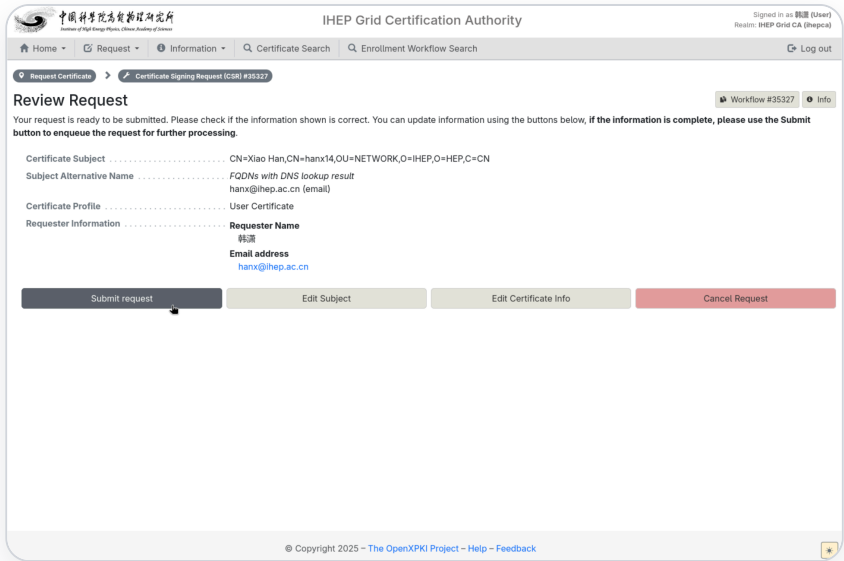
# Step 6 — Review and submit

**1** **Review Request**
check **Certificate Subject**, **SAN**, **Profile**, **Requester Information**; use **Edit Subject / Edit Certificate Info** to go back and fix anything.

**2** **Submit request**

**3** **Duplicate warning?**
Failed Policy Check means a repeated application — click **Submit with Policy Exception** and justify it (e.g. lost private-key password).

**Note**
Policy-exception requests show **Policy Violation Pending** and may take a little longer to be reviewed.




**Review Request** — the last chance to fix mistakes before Submit.

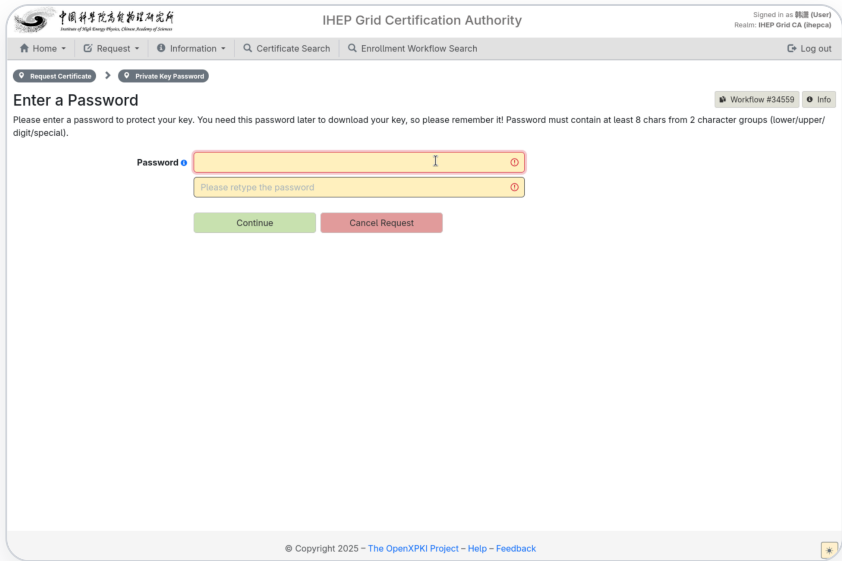
# Step 7 — Set the private key password

**1** **This is the Private Key Password**
It protects your private key — and you will type it **again when downloading** the certificate.

**2** **Rules**
At least 8 characters, mixing at least two of: upper case, lower case, digits, special characters.

**3** **Memorize it**
The CA cannot recover it. Losing it means re-applying via policy exception.

> **Note**
> Two different passwords exist: this **Private Key Password** (now), and an **Export Password** you will set when downloading the `.p12` (leave it empty to reuse this one). Neither is your LDAP password.

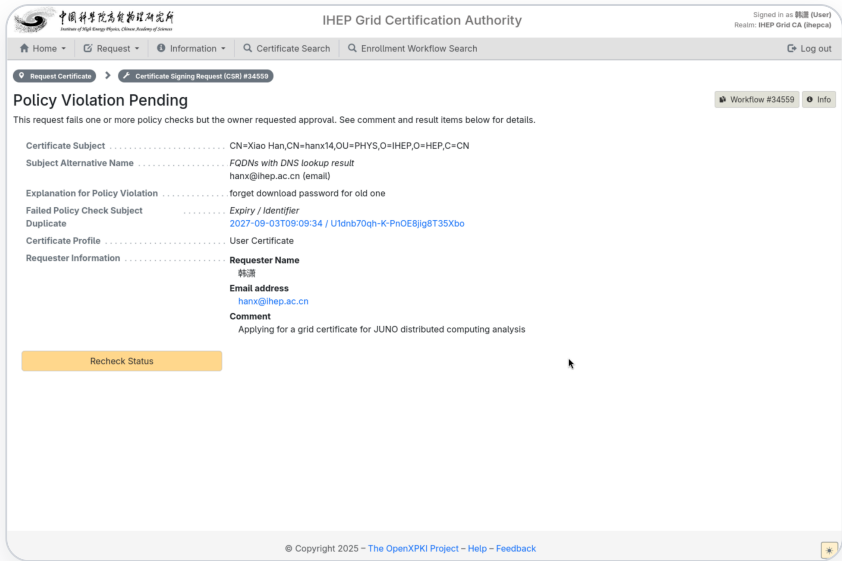



**Enter a Password** — the Private Key Password.

# Step 8 — Track your request

**Home → My Workflows** — click the workflow ID to see State, History and messages.

| Status | Meaning | Your action |
|---|---|---|
| `PENDING` / `ONHOLD` | Awaiting RA review | Wait; do **not** duplicate requests |
| `Policy Violation Pending` | Policy exception under review | Use **Recheck Status** to refresh |
| `SUCCESS` | Issued | Download from My Certificates |
| `REJECTED` | RA refused | Read the reason, fix, re-apply |
| `FAILURE` | Workflow error | Save the workflow ID, contact support |




After submission — waiting for RA review (here: a policy-exception request).

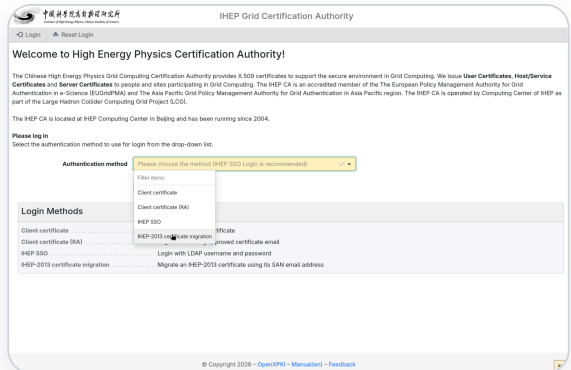
> ⚠ **Warning**
> The browser Back button does **not** cancel a request. Use the on-page **Cancel** action if you truly need to withdraw one.

# 4 · Migrate from IHEP-2013

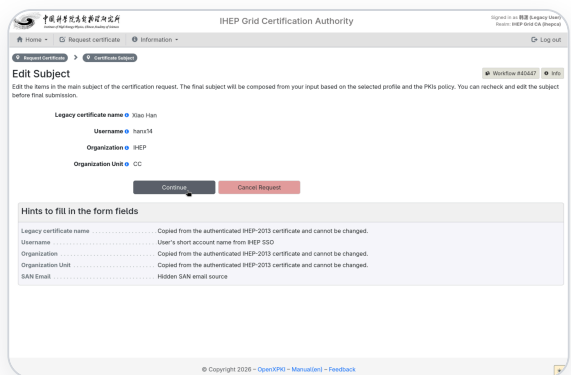
# Already have an old personal certificate?

**Migration entry = one-time shortcut** for users holding a valid, unrevoked IHEP-2013 **personal** certificate:

1. Install the old personal certificate **with its private key** into your browser first.
2. Open the WebUI — the browser pops up the **certificate selection**: pick your **personal** IHEP-2013 certificate.
3. In **Authentication method** choose **IHEP-2013 certificate migration** and click **Login**.
4. Identity fields are **auto-filled** from the old cert + LDAP — no re-typing. Set a password, submit, download.




The login page — choose **IHEP-2013 certificate migration**.




Subject **auto-filled** from the old certificate and LDAP — read-only.

# 5 · Use It on the Grid

# Download your PKCS#12

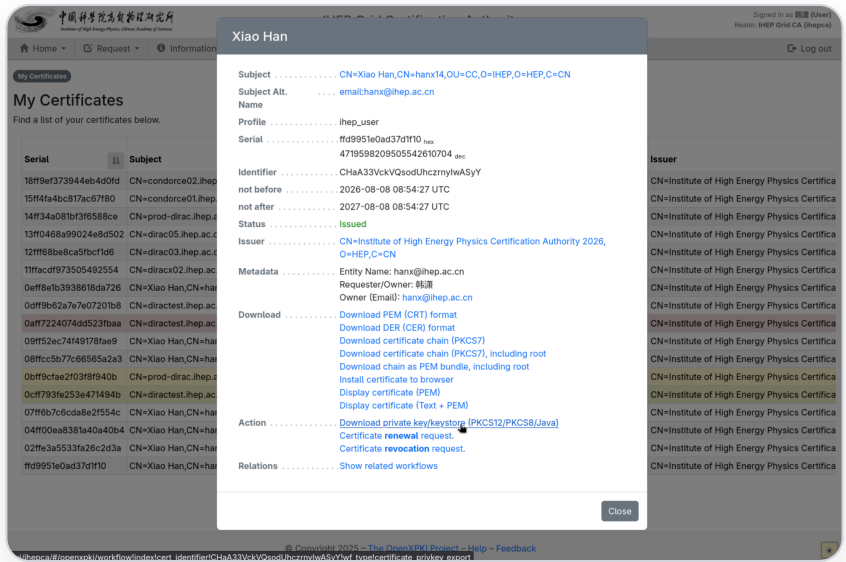
1. **Issued notification email**
   arrives with a direct link that opens your certificate.

2. **Home → My Certificates**
   click the certificate's Subject or serial number.

3. **Action → Download private key/keystore**

4. **Move the `.p12` somewhere safe**
   Out of the shared Downloads folder, restricted permissions. Not in /tmp, not on cloud drives.

> **Note**
> No download action? Either the private key is not held by the CA anymore, or the certificate came from EST — it does not mean the certificate is invalid.




Certificate detail — the **Action** area with the download entry.

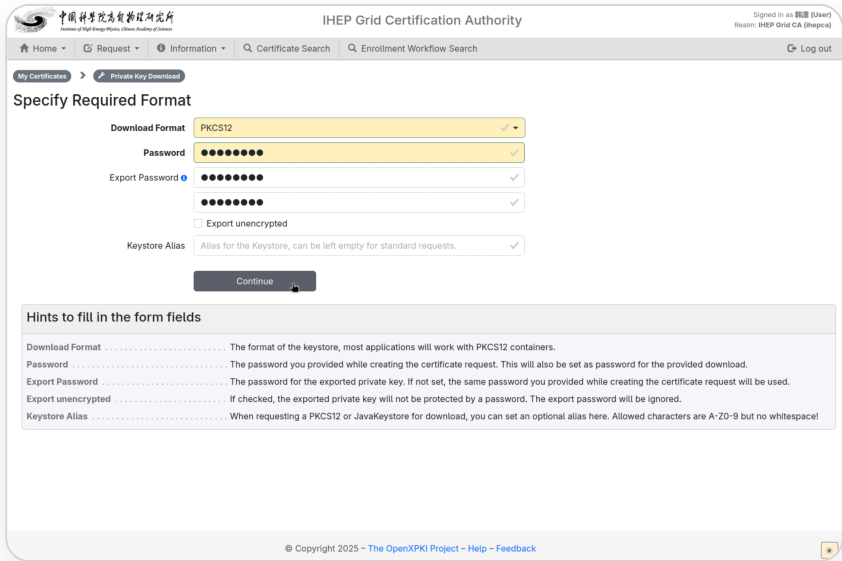
# Download: format and the two passwords

1. **Download Format: PKCS12**
   confirm it on the **Specify Required Format** page.

2. **Password**
   enter the **Private Key Password** you set in Step 7.

3. **Export Password**
   encrypts the downloaded `.p12` — leave empty to reuse the private key password.

4. **Continue → SUCCESS → Download**
   save the file **immediately**.




**Specify Required Format** — PKCS12, Private Key Password, Export Password.

> ⚠ **Warning**
> The download entry **disappears once you leave the SUCCESS page**. If you miss it, reopen the certificate and start the download action again.

# Install into Firefox

1. **Settings → Privacy & Security → Certificates → View Certificates**

2. **Your Certificates → Import**
   pick your `.p12`, enter the export password.

3. **Verify the entry**
   double-click it: **Issued By** = new IHEP Issuing CA; SAN contains your email.

4. **Try a client-cert website**
   pick your new certificate when the browser asks.




Firefox **Certificate Manager → Your Certificates → Import** — pick your `.p12` and enter the Export Password.

**Important**
Importing only a `.crt` / `.pem` (public part) does **not** work — the browser needs the private key, which lives in the `.p12`.

Tip: stick to one browser through apply → download → import, and avoid browser updates mid-workflow.

# Export PEM files for the grid CLI

Rename the downloaded `.p12` — it is usually named after you, e.g. `Xiao Han.p12` — to `myCert.p12`, and upload it to **KEKCC or whichever server you will use** (e.g. with `scp`). Grid tools expect `~/.globus/usercert.pem` + `userkey.pem`:

```
mkdir -p "$HOME/.globus" && chmod 700 "$HOME/.globus"

openssl pkcs12 -in myCert.p12 -clcerts -nokeys -out "$HOME/.globus/usercert.pem"
openssl pkcs12 -in myCert.p12 -nocerts -out "$HOME/.globus/userkey.pem"
chmod 644 "$HOME/.globus/usercert.pem"
chmod 600 "$HOME/.globus/userkey.pem"
```

**Verify what you got** (subject, issuer, dates, SAN, and key match):

```
openssl x509 -in "$HOME/.globus/usercert.pem" -noout -subject -issuer -dates -ext subjectAltName
test "$(openssl x509 -in "$HOME/.globus/usercert.pem" -noout -modulus | openssl sha256)" \
  = "$(openssl rsa -in "$HOME/.globus/userkey.pem" -noout -modulus | openssl sha256)" \
  && echo 'certificate and key match'
```

> ⚠ **Warning**
> Delete `myCert.p12` on that server once the conversion succeeds — don't leave copies of the encrypted key around. If `openssl pkcs12` complains about the format, drop or add `-legacy`. Never send the `.p12` to anyone to "fix" it.

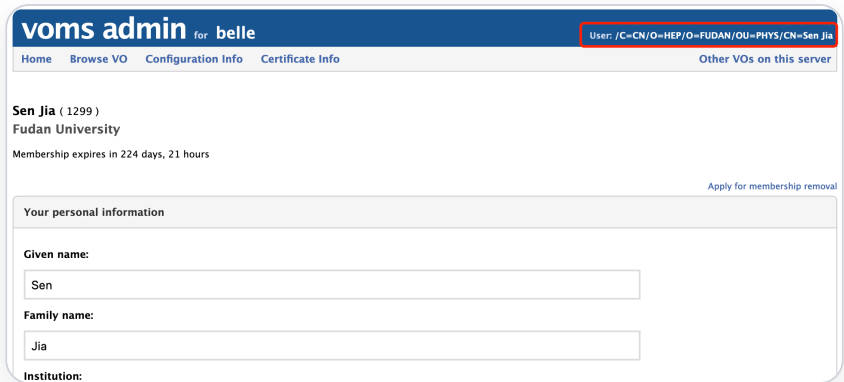
# Register your Belle II VO membership

1. **Open the Belle II VOMS page**
   voms.cc.kek.jp:8443/voms/belle — log in **with your new certificate** (your first real use of it!).

2. **Request membership**
   follow the page actions to register as a new member.

3. **Wait for approval**
   a VO admin confirms — usually within a day or two.

4. **Done**
   your DN now carries the `/belle` roles used by grid services.




**VOMS Admin for the belle VO** — log in with your certificate; your DN and membership appear on your profile page.

> ⚠ **Warning**
> Membership binds to your **DN**. A normal renewal keeps your DN — membership survives. But if your DN ever changes (different name spelling), you must register again.

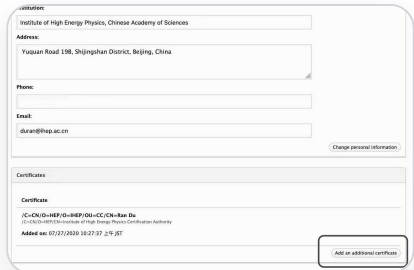
# New certificate: update your VOMS binding

1. **Log in with the old certificate**
   the one already bound to your VO account.

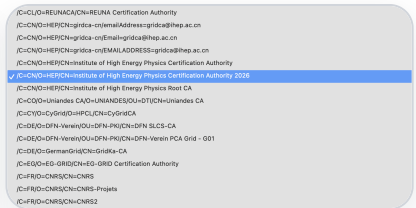
2. **Add an additional certificate**
   on your profile page, under **Certificates**.

3. **Select the new issuer**
   *Institute of High Energy Physics Certification Authority 2026* in the CA list.

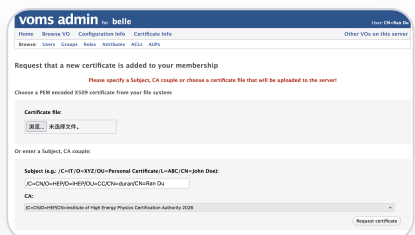
4. **Enter your new DN**
   slash form, e.g.
   `/C=CN/O=HEP/O=IHEP/OU=CC/CN=duran/CN=Ran Du` → **Request certificate**, then wait for VO admin approval.



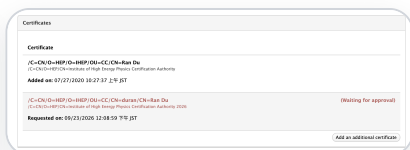

Profile → **Add an additional certificate**




Pick the **2026 issuer** in the CA list




Enter the new DN — or upload the **PEM only**




New certificate **waiting for approval**

**Important**
The form also accepts a **certificate file upload** — if you use it, upload the **PEM certificate only** (`usercert.pem`). **Never upload your `.p12` — it contains your private key.**

# Use it: gBASF2 proxy example

### On an IHEP login node (or KEKCC):

```
source /cvmfs/belle.kek.jp/grid/gbasf2/pro/bashrc
gb2_proxy_init
```

### What the proxy output shows:

```
subject      : /C=CN/O=HEP/O=IHEP/OU=CC/CN=hanx14/CN=Xiao Han
issuer       : /C=CN/O=HEP/CN=Institute of High Energy Physics
               Certification Authority 2026
timeleft     : 23:53:58
DIRAC group  : belle
VOMS         : True
VOMS fqan    : ['/belle']
```

> **Note**
> The proxy is a **short-lived credential** derived from your certificate. Your long-lived secret (private key) never leaves your machine.

### Remember:

- Certificate expires in ~1 year — renew in time (§6)
- New certificate ⇒ new proxy: re-run `gb2_proxy_init`
- Belle II VO membership (VOMS `/belle`) is registered against your DN — changing your DN breaks it

# gbasf2: setup and your first job

### Setup (KEKCC or IHEP login node):

```
source /cvmfs/belle.kek.jp/grid/gbasf2/pro/bashrc
gb2_check_release   # check your gbasf2 release
```

- ⓘ **basf2 and gbasf2 are NOT compatible** — use separate shell sessions
- gbasf2 and basf2 versions are **independent** of each other

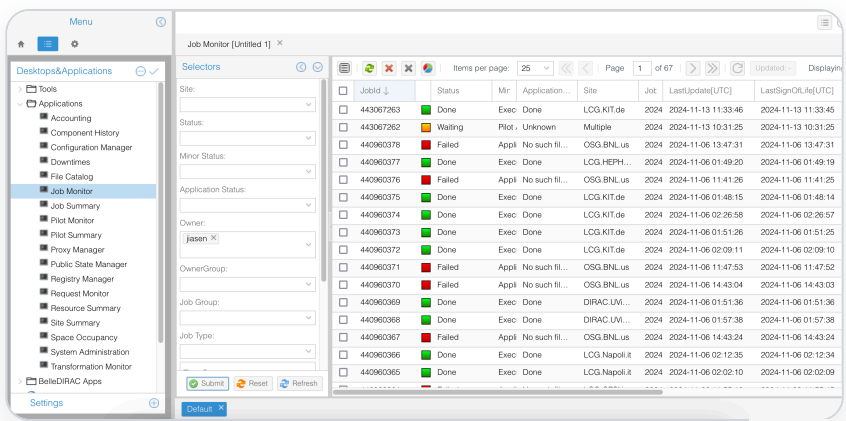
### Golden rule

> **Important**
> Always run your script with **basf2 first** on local samples — only then submit it to the grid.

### Submit a job — same logic as basf2:

```
gbasf2 ./ana01.py -s release-08-01-05 \
  -p 20261114_myproject --input_dslist input1.txt
```

- `-s` basf2 release · `-p` your project name · `--input_dslist` input dataset
- Monitor jobs in the **DIRAC portal**




The BelleDIRAC **Job Monitor**

# Useful commands and where to get help

| Command | Purpose |
|---|---|
| `gb2_job_status -p PROJ` | Job status of your project |
| `gb2_job_output -p PROJ --status Failed` | Fetch logs of failed jobs |
| `gb2_job_reschedule -p PROJ` | Retry failed jobs |
| `gb2_job_kill -p PROJ` | Kill remaining jobs |
| `gb2_ds_list /belle/user/<you>/PROJ` | Count output files |
| `gb2_ds_get /belle/user/<you>/PROJ` | Download outputs |
| `gb2_ds_rm "/belle/user/<you>/*"` | Remove obsolete datasets |

### Questions & discussion

- questions.belle2.org — the computing Q&A forum

### Mailing list

- comp-users-forum@belle2.org — user discussion; subscribe at the list page

### Documentation

- Belle II computing documentation — online book (BASF2 / gbasf2 / grid)

> **Note**
> For **certificate** problems (login, issuance, renewal, revocation) — contact the IHEP CA support, not the Belle II forum.

# 6 · Renewal & Revocation

# Renew before expiry

1. **My Certificates → open a still-valid certificate**

2. **Action → Renew**
   profile and Subject are carried over — verify, don't retype.

3. **New export password → Submit**

4. **Download the new `.p12`, reinstall**
   browser + `.globus`, then make a fresh proxy.

**Good to know:**

- Renewal issues a **new private key and serial** — never stitch the new cert onto the old key by hand
- The old certificate stays valid until its original expiry → smooth switchover
- Renew ≠ change: different department, name spelling or ownership = **new application**, not renewal
- Expiry reminder emails are sent automatically — keep your LDAP email correct

# Revoke when compromised

**When to revoke immediately:**

- Private key or `.p12` suspected leaked
- You stop being the responsible person (Robot/Host)

**How:**

1. **My Certificates → open the certificate**
2. **Action → Revoke**
   choose the most accurate reason, explain for the RA.
3. **Submit**
   and wait for CRL publication.

> **Important**
> **Revocation is irreversible.** If the key may be compromised: stop using it first, then revoke — do not wait for expiry, and do not just delete local files (copies may exist elsewhere).

**Propagation:** relying parties fetch the CRL on their own schedule — a revoked cert may still be accepted for a while by some services.

# 7 · Robot & Host Certificates

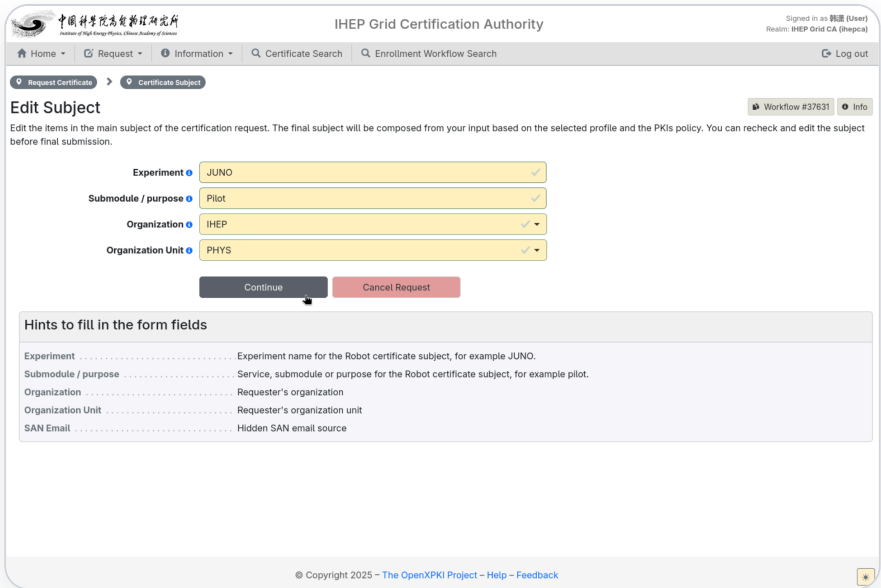
# Robot certificates

For **non-human identities**: experiment pilots, production automation, service agents — not for natural persons.

**Application highlights (WebUI, like §3):**

- **Experiment** — e.g. `Belle` (letters, digits, dot, underscore, hyphen)
- **Submodule / purpose** — e.g. `pilot`
- **OU / O** — the responsible group
- SAN email = the responsible person's LDAP email (automatic)

> **Important**
> Every Robot certificate needs a **responsible person**. Ownership change, purpose change or key leak ⇒ revoke and re-apply — do not "migrate" a Robot cert to someone else.




**Robot certificate** — Experiment + Submodule / purpose; the second CN is fixed to **Robot**.

## Resulting DN shape:

```
1   CN=Belle pilot,CN=Robot,OU=<dept>,O=<inst>,O=HEP,C=CN
```

# Host certificates & EST

**WebUI first issue (site admin):**

- **Hostname** = primary FQDN (`host01.ihep.ac.cn`, never short names)
- **Additional hostnames** = one FQDN per entry, no wildcards, no IPs/URLs
- Names are issued lowercase; email SAN = applicant's LDAP mail

**Install to the host** (`.p12` → PEM):

```
openssl pkcs12 -in host01.p12 -clcerts -nokeys \
  -out /etc/grid-security/hostcert.pem
openssl pkcs12 -in host01.p12 -nocerts -nodes \
  -out /etc/grid-security/hostkey.pem
chmod 444 /etc/grid-security/hostcert.pem
chmod 400 /etc/grid-security/hostkey.pem
```

**EST auto-renewal (hands-free afterwards):**

- Script `ihep-host-auto.sh` on the host
- Renews automatically within **45 days** of expiry
- Old IHEP-2013 host certs migrate through the same script (mTLS, one-time per hostname)
- Site admins deploy it — users normally never touch this

> **Note**
> Full details (wrapper script, cron, RELOAD_CMD) in the **IHEPCA User Guide** — linked at the end.

# 8 · Troubleshooting & Security

# Common problems

| Symptom | Check first |
|---|---|
| WebUI login fails | LDAP account/password; wrong client cert selected; system clock |
| Cannot find my certificate | **My Certificates** + which email you logged in with |
| Stuck in `PENDING` | workflow ID from **My Workflows** → contact RA; don't re-submit |
| No "download key" action | Key not CA-generated / already exported / EST-issued — not an error |
| Browser never offers the certificate | Import the `.p12` (with key!); check cached choices |
| Certificate chain fails | Client must trust the **new Root + Issuing CA**; CRL URLs reachable |

**Note**

Contact support with: certificate serial or Subject · workflow ID · time (UTC) · full error text. **Never** send private keys, `.p12`, or passwords.

# Security habits

### Move `.p12` out of Downloads immediately

Restricted location and permissions. Not /tmp, not sync drives, not group-readable home.

### One export password, kept secret

Not the LDAP password; never recovered by the CA; never shared.

### Revoke on suspicion

Lost key or leaked file → stop using it and revoke. Fast beats perfect.

### Keep your LDAP email current

Expiry alerts and ownership follow that address.

### Never

e-mail / chat / cloud-share a `.p12`, `userkey.pem` or password

### Never

write passwords or key material in the Comment field

# Summary — do it this week

1. **Check your IHEP Account login and LDAP email**
   — everything hangs on these two.

2. **Request / migrate your certificate**
   at gridca.ihep.ac.cn — new request, or the IHEP-2013 migration entry.

3. **Install to browser + `.globus`**
   — bring a working proxy to the hands-on session.

4. **Register / update your VOMS membership**
   join the belle VO — or add your new DN.

**Important**
**Deadline:** old-CA certificates stop being recognized at the **end of 2026**.

**IHEP CA WebUI**
gridca.ihep.ac.cn/webui/ihepca

**IHEP CA user guide**
cchelp.ihep.ac.cn/grid/new-ca

**gbasf2 user manual**
Belle II online book — gbasf2

**Belle II VOMS**
voms.cc.kek.jp/voms/belle

**Questions & forum**
questions.belle2.org

**DIRAC portal**
dirac.cc.kek.jp/DIRAC



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| **Acknowledgements** | The Interoperable Global Trust Federation (IGTF) · The Asia Pacific Grid Policy Management Authority (APGridPMA) · The European Policy Management Authority for Grid Authentication (EUGridPMA) |
| **WebUI** | gridca.ihep.ac.cn |

Questions?

# Thank you!